Lecture 2

Stochastic Analysis on Hilbert Spaces

1.2. Wiener Processes

Let Q be as in the previous section and let $(U, \langle \cdot, \cdot \rangle_U)$ be any separable, real Hilbert space.

Definition 1.12. A U-valued stochastic process $W(t)$, $t \in [0,T]$, on $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **(standard) Q-Wiener process**, if

(i) $W(0) = 0$,
(ii) $t \mapsto W(t)$ is $\mathbb{P}$-a.s. continuous,
(iii) for $0 = t_0 < t_1 < \ldots < t_n \leq T$ the increments

$$W(t_i) - W(t_{i-1}), \quad i = 1, \ldots, n$$

are independent, $\mathcal{N}(0, (t_i - t_{i-1})Q)$-distributed.

Proposition 1.13 (canonical representation of Q-Wiener processes). *Let $(e_k)_{k \geq 1}$ be a CONS, consisting of eigenfunctions of Q. Then $W(t)$, $t \in [0,T]$, is a Q-Wiener process, if and only if*

$$W(t) = \sum_{k=1}^{\infty} \sqrt{\lambda_k} \beta_k(t) e_k, \quad t \in [0,T], \tag{1.21}$$

where $\{\beta_k(t) : \lambda_k > 0\}$ are independent, 1-dimensional (continuous) Brownian motions. The sequence (1.21) converges $\mathbb{P}$-a.s. on $C([0,T];U)$ and in $L^p(\Omega, \mathcal{F}, \mathbb{P}; C([0,T];U))$ for all $p < \infty$, i.e.

$$\lim_{n \to \infty} \mathbb{E}\Big(\sup_{t \in [0,T]} \| \sum_{k=1}^{n} \sqrt{\lambda_k} \beta_k(t) e_k - W(t) \|_U^p \Big) = 0.$$

Sketch of proof:
We only present the proof for the case $p = 2$. The general case and the $\mathbb{P}$-a.s. convergence follows similarly from generalizations of the martingale convergence theorems to Hilbert-space valued martingales. To simplify notations let

$$W^n(t) := \sum_{k=1}^{n} \sqrt{\lambda_k} \beta_k(t) e_k.$$

Then

$$\|W^n(t) - W(t)\|_U^2 = \| \sum_{k=n+1}^{\infty} \sqrt{\lambda_k} \beta_k(t) e_k \|_U^2 = \sum_{k=n+1}^{\infty} \lambda_k \beta_k^2(t).$$

Hence

$$\begin{aligned}\mathbb{E}\left(\sup_{t\in[0,T]}\|W^n(t)-W(t)\|_U^2\right) &= \mathbb{E}\left(\sup_{t\in[0,T]}\sum_{k=n+1}^{\infty}\lambda_k\beta_k^2(t)\right)\\ &\leq \sum_{k=n+1}^{\infty}\lambda_k\,\underbrace{\mathbb{E}\left(\sup_{t\in[0,T]}\beta_k^2(t)\right)}_{\leq C_T(\text{ Doob})}\\ &\leq C_T\sum_{k=n+1}^{\infty}\lambda_k \overset{n\to\infty}{\longrightarrow} 0\end{aligned}$$

Conversely: If $W(t)$ is a Q-Wiener process, define

$$\beta_k(t) := \begin{cases}\frac{1}{\sqrt{\lambda_k}}\langle W(t), e_k\rangle & \text{, if } \lambda_k > 0\\ 0 & \text{, otherwise.}\end{cases}$$

It then follows for $s \leq t$ and $k \geq 1$ that

$$\beta_k(t)-\beta_k(s) = \frac{1}{\sqrt{\lambda_k}}\underbrace{\langle\underbrace{W(t)-W(s)}_{\sim\mathcal{N}(0,(t-s)Q)}, e_k\rangle}_{\sim\mathcal{N}(0,(t-s)\lambda_k)},$$

so that $\beta_l(t), k \geq 1$ are 1-dimensional Wiener processes, hence 1-dimensional continuous Brownian motions. To see that they are indeed independent let $k \neq l$ and $s \leq t$. Then:

$$\begin{aligned}\mathrm{Cov}(\beta_k(t),\beta_l(s)) &= \frac{1}{\sqrt{\lambda_k\lambda_l}}\mathbb{E}\left(\langle W(t),e_k\rangle\langle W(s),e_l\rangle\right)\\ &= \frac{1}{\sqrt{\lambda_k\lambda_l}}\left[\mathbb{E}\left(\langle W(t)-W(s),e_k\rangle\langle W(s),e_l\rangle\right)+\mathbb{E}\left(\langle W(s),e_k\rangle\langle W(s),e_l\rangle\right)\right]\\ &= \frac{1}{\sqrt{\lambda_k\lambda_l}}\left[\underbrace{\mathbb{E}\left(\langle W(t)-W(s),e_k\rangle\right)}_{=0}\underbrace{\mathbb{E}\left(\langle W(s),e_l\rangle\right)}_{=0}+s\langle Qe_k,e_l\rangle\right]\\ &= \frac{1}{\sqrt{\lambda_k\lambda_l}}s\lambda_k\underbrace{\langle e_k,e_l\rangle}_{=0}\\ &= 0\end{aligned}$$

□

Definition 1.14 (Q-Wiener process w.r.t. some filtration $(\mathcal{F}_t)$). An U-valued stochastic process $W(t)$, $t \in [0,T]$, is called Q-Wiener process w.r.t. a given filtration $(\mathcal{F}_t)$, if

(i) $W(t)$ is $\mathcal{F}_t$-measurable, i.e. W is $(\mathcal{F}_t)$-adapted,
(ii) $t \mapsto W(t)$ is $\mathbb{P}$-a.s. continuous,
(iii) $W(t)-W(s)$ is independent of $\mathcal{F}_s$ for all $0 \leq s < t \leq T$ and $N(0,(t-s)Q)$-distributed.

Similar to the finite-dimensional case it holds that a Q-Wiener process $W(t)$, $t \in [0,T]$, always is a Q-Wiener process w.r.t. the minimal right-cotinuous filtration

$$\mathcal{F}_t^W = \bigcap_{s>t}\mathcal{F}_s^0, \quad \mathcal{F}_s^0 = \sigma(W(r) : r \in [0,s])$$

generated by W.

1.3. Stochastic Integration on Hilbert spaces

The aim of this section is to define the stochastic-integral

$$\int_0^t \Phi(s)dW(s)$$

w.r.t. some H-valued Q-Wiener process.

1.3.1. Martingales in Banach spaces. Let $(E, \|\cdot\|)$ be a (real) Banach space, i.e. a complete normed ($\mathbb{R}$-) vector space. Let μ be a finite measure on a measure space $(\Omega, \mathcal{A})$.

1.3.1.1. *Bochner-Integral.* (E-valued μ-integration)

We want to extend the Lebesgue integral

$$\int_\Omega f d\mu \in \mathbb{E}$$

for measurable real-valued functions $f : \Omega \longrightarrow \mathbb{R}$ to suitable sets of measurable mappings $f : \Omega \longrightarrow E$.

Similar to the case of the Lebesgue-integral we first consider elementary E-valued step functions

$$f = \sum_{k=1}^n x_k 1_{A_k} \tag{1.22}$$

with $x_k \in E$ and $A_k \in \mathcal{A}$. W.l.o.g. we may assume that A_k are pairwise disjoint.

For such integrands we then define the integral as follows:

$$\int_\Omega f\, d\mu := \sum_{k=1}^n x_k \mu(A_k) \in E,$$

and similar to the case of the Lebesgue-integral one can show that this definition is indepedent of the given representation of the elementary step function f.

It is then easy to see that the following **Bochner inequality** holds:

$$\left\| \int_\Omega f d\mu \right\|_E \leq \int_\Omega \|f\|_E d\mu.$$

Indeed,

$$\begin{aligned}
\left\| \int_\Omega f d\mu \right\|_E &= \left\| \sum_{k=1}^n x_k \mu(A_k) \right\|_E \leq \sum_{k=1}^n \|x_k\|_E \mu(A_k) \\
&= \sum_{k=1}^n \|x_k\|_E \int_\Omega 1_{A_k}\, d\mu = \int_\Omega \sum_{k=1}^n \|x_k\|_E 1_{A_k}\, d\mu \\
&= \int_\Omega \|f\|_E d\mu.
\end{aligned}$$

Using the Bocher inequality we can extend the integral $\int f d\mu$ to all integrands

$$\begin{cases} f & : \Omega \longrightarrow E \text{ satisfying} \\ & \bullet\ f \text{ is } \textbf{strongly measurable}, \text{ i.e., } f = \lim_{n\to\infty} f_n \ \mu - a.e. \\ & \qquad\qquad f_n \text{ is of type (1.22)} \\ & \bullet\ \int \|f\|_E d\mu < \infty \end{cases} \tag{1.23}$$

Remark For mappings $f : \Omega \longrightarrow E$ taking values in an infinite dimensional normed vector-space E, it is not true in general, whether $\mathcal{A}/\mathcal{B}(E)$-measurability implies that it can be approximated by pointwise limits elementary step functions. $\mathcal{A}/\mathcal{B}(E)$-measurability of a mapping f implies that f is **weakly measurable** in the sense that for all continuous linear functionals $\ell \in E'$ the real-valued mapping $\ell(f)$ is $\mathcal{A}/\mathcal{B}(\mathbb{R})$-measurable. It is then a classical result in functional analysis, called **Pettis theorem**, that the following holds true:

THEOREM 1.15 (Pettis theorem). *A necessary and sufficient condition that $f : \Omega \longrightarrow E$ is (strongly) measurable is that it is weakly measurable and separably valued.*

Here, **separably valued** means that the range $f(\Omega)$ of the mapping f is a separable subset of E. It is obvious that if E is separable, e.g. $E = \mathbb{R}^d$, $L^p(\mathbb{R}^d)$, l^p, ..., then every (weakly) measurable function $f : \Omega \longrightarrow E$ already is strongly measurable.

The set of all admissible integrands f will be denoted by $\mathcal{L}^1(\Omega, \mathcal{A}, \mu; E)$ or simply with $\mathcal{L}^1(\mu; E)$. Let $L^1(\mu; E)$ the set of all μ-equivalence classes in $\mathcal{L}^1(\mu; E)$, w.r.t. the equivalence relation $f \sim g$ if and only if $f = g$ μ-a.e. Similarly we define $\mathcal{L}^p(\mu; E)$ and $L^p(\mu; E)$.

Properties of the Bochner integral

(i) **Bochner inequality**

$$\left\| \int_\Omega f d\mu \right\|_E \leq \int_\Omega \|f\|_E d\mu$$

(ii) **Linearity** Let $L : E \longrightarrow F$ be linear and continuous, where F is another Banach space. Then:

$$L\left(\int_\Omega f\, d\mu\right) = \int_\Omega L \circ f\, d\mu.$$

In particular: $\alpha, \beta \in \mathbb{R}$, $f = (f_1, f_2) \in E \times E$,

$$L : E \times E \longrightarrow E, \quad (x_1, x_2) \mapsto \alpha x_1 + \beta x_2.$$

Then

$$\begin{aligned}\alpha \int_\Omega f_1 d\mu + \beta \int_\Omega f_2 d\mu &= L\left(\int_\Omega f d\mu\right)\\ &= \int_\Omega L \circ f d\mu\\ &= \int_\Omega \alpha f_1 + \beta f_2 d\mu.\end{aligned}$$

(iii) **Fundamental theorem of calculus**

Let $f \in C^1([a,b];E)$, i.e., differentiable curve in E. Then

$$f(t) - f(s) = \int_s^t f'(r)\, dr$$

for all $a \le s \le t \le b$.

1.3.1.2. *E-valued conditional expectations.* Form now on we assume that E is a separable Banach space. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. We then have the following result on the existence of E-valued conditional expectations:

THEOREM 1.16. *Let $X : \Omega \longrightarrow E \in \mathcal{L}^1(\Omega, \mathcal{A}, \mathbb{P}; E)$ be a Bochner integrable E-valued random variable and $\mathcal{A}_0$ be a sub-σ-algebra of $\mathcal{A}$. Then there exists a $\mathbb{P}$-a.s. uniquely determined Bochner integrable E-valued random variable $X_0 \in \mathcal{L}^1(\Omega, \mathcal{A}_0, \mathbb{P}; E)$ such that*

$$\int_A X_0\, d\mathbb{P} = \int_A X\, d\mathbb{P} \qquad \forall\, A \in \mathcal{A}_0.$$

The $\mathcal{A}_0$-measurable random variable X_0 is called the conditional expectation of X given $\mathcal{A}_0$ and we write

$$X_0 =: \mathbb{E}\left(X \mid \mathcal{A}_0\right).$$

Moreover, Bochner's inequality holds

$$\|E\left(X \mid \mathcal{A}_0\right)\|_E \le E\left(\|X\|_E \mid \mathcal{A}_0\right).$$

The proof of the theorem is first reduced to the case of elementary step functions $X = \sum_k x_k 1_{A_k}$ with A_k, $k \ge 1$, pairwise disjoint. In this case clearly,

$$E\left(X \mid \mathcal{A}_0\right) = \sum_k x_k E\left(1_{A_k} \mid \mathcal{A}_0\right),$$

and using the triangle inequality we obtain that

$$\begin{aligned}\|E\left(X \mid \mathcal{A}_0\right)\|_E &= \left\|\sum_k x_k E\left(1_{A_k} \mid \mathcal{A}_0\right)\right\|_E\\ &\le \sum_k \|x_k\|_E E\left(1_{A_k} \mid \mathcal{A}_0\right) = E\left(\sum_k \|x_k\|_E 1_{A_k} \mid \mathcal{A}_0\right) = E\left(\|X\|_E \mid \mathcal{A}_0\right).\end{aligned}$$

The general case can then be obtained considering pointwise limits of random variables of the above type.

EXAMPLE 1.17. Let $\mathcal{A}_0 := \sigma\left(A_k : k = 1, ..., n\right)$, A_k pairwise disjoint. Then

$$\mathbb{E}(X|\mathcal{A}_0) = \sum_{k=1}^{n} \frac{1}{\mu(A_k)} \int\limits_{A_k} X \, d\mathbb{P} \, 1_{A_k}$$

(with the convention $\frac{1}{\mu(A_k)} \int\limits_{A_k} X \, d\mathbb{P} := 0$, if $\mu(A_k) = 0$).

We can now extend the definition of a real-valued martingale $(M_t)_{t\geq 0}$ w.r.t. a given filtration $(\mathcal{A}_t)_{t\geq 0}$ to the general case of E-valued martingales.

DEFINITION 1.18. An E-valued stochastic process $(M_t)_{t\geq 0}$ on $(\Omega, \mathcal{A}, \mathbb{P})$ is called an $(\mathcal{A}_t)$-martingale, if

(i) $\mathbb{E}(\|M_t\|) < \infty$ for all $t \geq 0$,
(ii) M_t is $\mathcal{A}_t$-measurable for all $t \geq 0$,
(iii) $\mathbb{E}(M_t|\mathcal{A}_s) = M_s$ for all $0 \leq s \leq t$.

Remark It is an immediate consequence of the definition that (M_t) is an $(\mathcal{A}_t)$-martingale if and only if for all $l \in E'$ the real-valued process $(l(M_t))$ is a real-valued $(\mathcal{A}_t)$-martingale.

EXAMPLE 1.19. (i) $X \in \mathcal{L}^1(\mathbb{P}; E)$, then $X_t = \mathbb{E}(X|\mathcal{A}_t)$ is a martingale.

(ii) Consider discrete time $t = 0, 1, 2, ...$ and $Y_1, Y_2, ... \in \mathcal{L}^1(\mathbb{P}; E)$, iid and $\mathcal{A}_t = \sigma(\{Y_1, ..., Y_t\})$. Then

$$X_t = \sum_{s=1}^{t} (Y_s - \mathbb{E}(Y_s))$$

is a martingale.

(iii) As a generalization of the previous example to continuous time consider a U-valued Q-Wiener process (W_t) w.r.t. $(\mathcal{A}_t)$ (where U is a separable real Hilbert space). Then (W_t) is an $\mathcal{L}^2$-integrable continuous martingale with

$$\mathbb{E}(\|W_t\|_U^2) = t \ \mathrm{tr}(Q).$$

Indeed, similar to the case of finite dimensional Brownian motion,

$$\begin{aligned}
\mathbb{E}(W_t|\mathcal{A}_s) &= \mathbb{E}(W_t - W_s + W_s|\mathcal{A}_s) \\
&= \underbrace{\mathbb{E}(W_t - W_s|\mathcal{A}_s)}_{=\mathbb{E}(W_t - W_s)=0} + \underbrace{\mathbb{E}(W_s|\mathcal{A}_s)}_{=W_s} \\
&= W_s,
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}(\|W_t\|_U^2) &= \mathbb{E}\left(\|\sum_{k=1}^{\infty} \sqrt{\lambda_k} \beta_k(t) e_k\|_U^2\right) = \mathbb{E}\left(\sum_{k=1}^{\infty} \lambda_k \beta_k^2(t)\right) \\
&= \sum_{k=1}^{\infty} \lambda_k \mathbb{E}\left(\beta_k^2(t)\right) = t \sum_{k=1}^{\infty} \lambda_k \\
&= t \ \mathrm{tr}(Q).
\end{aligned}$$

Remark The notion of sub- and super-martingales cannot be immediately generalized to arbitrary Banach spaces, unless we assume that we are given some partial ordering "$\leq$" which is for example the case on L^p-spaces. We will not use this additional structure in the infinite-dimensional case and therefore do not study sub- and super-martingales in more detail.

Similar to the case of the real-valued conditional expectation we have:

THEOREM 1.20 (Doob's maximal inequality). *Let (M_t) be a right-continuous $(\mathcal{A}_t)$-martingale. Then:*

$$\left(\mathbb{E}\left(\sup_{0\leq t\leq T}\|M_t\|^p\right)\right)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\left(\|M_T\|^p\right)\right)^{\frac{1}{p}}, \quad \forall p>1.$$

Main consequence: Let (M_t^n), $n\geq 1$, be an $\mathcal{L}^p$-Cauchy sequence of continuous martingales, i.e., $\lim\limits_{n,m\to\infty}\left(\mathbb{E}\left(\|M_T^n - M_T^m\|_E^p\right)\right)^{\frac{1}{p}} = 0$ for some $p>1$. Then there exists a continuous martingale (M_t) (w.r.t. the same filtration), such that

$$\lim_{n\to\infty}\mathbb{P}\left(\sup_{0\leq t\leq T}\|M_t^n - M_t\|_E \geq \varepsilon\right) = 0 \quad \forall\varepsilon>0\,,$$

hence $\lim\limits_{n\to\infty} M_t^n = M_t$ uniformly on $[0,T]$ in probability.

In other words:

$$\mathcal{M}_T^p := \{(M_t)_{t\in[0,T]}\ E\text{-valued, cont. } (\mathcal{A}_t)-\text{martingale},$$
$$\|M\|_{\mathcal{M}_T^p} := \sup_{0\leq t\leq T}\left(\mathbb{E}\left(\|M_t\|_E^p\right)\right)^{\frac{1}{p}} < \infty\}$$

is (again) a Banach space (if $p=2$ a Hilbert space).